Andromeda contains a gigantic black hole, 100 million times heavier than the Sun and as big across as the Earth's orbit (the same weight and size as *Interstellar*'s Gargantua; Chapter 6). It resides in the middle of the central bright sphere in Figure 2.2.

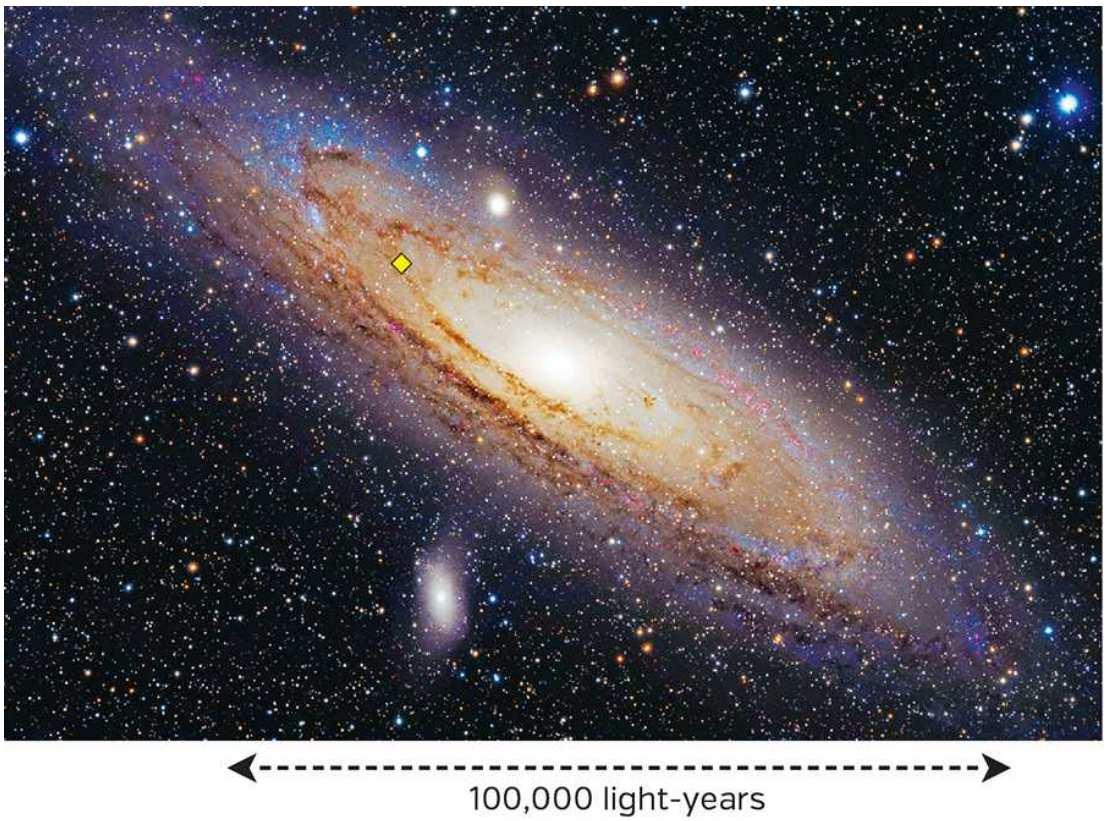


Fig. 2.2. The Andromeda galaxy.

Solar System

Stars are large, hot balls of gas, usually kept hot by burning nuclear fuel in their cores. The Sun is a fairly typical star. It is 1.4 million kilometers across, about a hundred times larger than the Earth. Its surface has flares and hot spots and cooler spots, and is fascinating to explore through a telescope (Figure 2.3).

Eight planets, including the Earth, travel around the Sun in elliptical orbits, along with many dwarf planets (of which Pluto is the most